## Crafting a Program for Counterfoil Design Research

What do you imagine when you think of a *convivial* world? For the philosopher Ivan Illich, conviviality meant "individual freedom realized in personal interdependence". It meant a world that enabled "all people to define the images of their own future" using tools that support meaningful and responsible work. In his book *Tools for Conviviality* [1], Illich identified a series of industrial threats to a convivial life, where the development of new technologies leads to the concentration of power and profit in the hands of very few, increased surveillance and control, forced consumption, social polarization, and biological degradation. Over the past fifty years, these issues have only intensified the multiple social, environmental, and political crises we are now facing [2].

Illich's response to these crises took the form of Counterfoil research, named for how it ran counter to the dominant forms of research in service of industry [3]. Its goal was to identify when and how systems became inimical to conviviality, and then "to devise tools and tool systems that optimize the balance of life, thereby maximizing liberty for all" [1]. In the face of radical computing-driven changes that seek to alter the way we live and work, often for the worse, I propose to develop methods for Counterfoil Design Research by building on my prior work, and learning from communities—both within and outside academia—that take a critical and intentional approach to design, one that "commits to a possible, by means of resisting the probable" [4].

## From Friend to Foil

If design is "the interaction between understanding and creation" [5], then my research journey has been one where a more critical understanding has over time caught up with ideas of creation. Around the start of my PhD, Spatial Computing through headsets and in the "metaverse" was hailed as the "next evolution in social technology" [6]. For all its promise, it was clear to me that headsets in their current form were very far from widespread availability and adoption. My dissertation focuses on designing for the "Everyday Spatial Computer" [7] by maximizing what we can achieve with phones, tablets, and laptops. Instead of giving in to the seamlessness of the metaverse, I developed new mobile-based Augmented Reality (AR) interfaces [8] that enable people to meaningfully collaborate with each other across space and over time as they navigate the seamful trajectories [9] of everyday life—a conceptual device that builds upon key research in Human-Computer Interaction (HCI) [10, 11, 12]. I demonstrated how these interfaces enable rich, embodied interactions with people and places through studies both in the lab and in the real world [8, 13]. Across these projects, I used design to question the exclusionary sociotechnical imaginaries [14] and oppressive visions [15] of the dominant headset-based approach, and move towards an alternate vision of spatial computing that works for all of us.

The collective attention of industry and academia has recently shifted away from spatial computing, and towards Artificial Intelligence (AI). During an internship at Microsoft Research in 2024, I explored how group work over the course of a project might be supported with Generative AI. I began by understanding the importance of *Temporal Work* [16] by learning from the fields of organizational science and cognitive science. I conducted a qualitative analysis of conversations across multiple series of real meetings, and created a design framework for interfaces seeking to support temporal work. I then used this framework to design a range of interface concepts informed by the extent to which AI models can process temporal information. A key contribution of this project was to show the value of an approach to design that is grounded, not in the capabilities of ever-changing AI models, but in an understanding of how real work actually takes place [17].

The world has changed since the summer of 2024, as has my stance as a designer. Today, it is crucial that we design tools and tool systems that serve as foils to the status quo and help realize visions of a convivial future.

## The Components of a Counterfoil

Effecting real change is not merely about tools and technologies. We need to address the "project" of technology as a whole, and avoid being caught up in tackling decoys that shift our attention away from the material, political, and sociotechnical challenges at hand [18]. However, in important ways, it *is* about the tools and technologies we use in the everyday. At a time when technologies—in particular those associated with AI—are being wielded to increase the precarity of work and life for those outside the capitalist class [19, 20, 21], the tools we use and the reasons behind their use are key sites of leverage, both for continued control, and the possibility of change. Returning to *Tools for Conviviality*, Illich warns against the creation of Radical Monopolies where "a major tool rules out natural competence ... imposes compulsory consumption and thereby restricts personal autonomy". When considering the worst implementations of AI, we may well be facing the beginnings of a radical monopoly over human interaction and thought. Today, we still have some "latitude of choice" to take a different trajectory [22] and it is essential that we try to change course. Reflecting on my prior work, I have found it helpful to frame Counterfoil research as using design to give us **(1) Reasons to Think**, about alternate approaches through the creation of new tools [8] and design methods [17], **(2) Reasons to Turn**, away from the dominant trajectory of development by connecting research more strongly to our experiences in the everyday [13] and **(3) Reasons to Try**, and keep moving towards visions of technology that work for all of us [9].

## Why Craft a Program?

This initial framing of Counterfoil research in terms of design is by no means a new idea. For decades and across the allied fields of HCI, Science and Technology Studies, Sociology, and more, countless designers, researchers, activists, and communities have come together to do work that insists on better possible futures [4]. Learnings from across these efforts have been synthesized into frameworks such as Design Justice [23] and Just Sustainability Design [24]. The work that I am proposing here is similarly biased towards action in the real world, and aims to follow in this long tradition of "crafting" forms of resistance in two key directions:

**1. Designing Counterfoil Infrastructure:** I will study, design, and share interfaces that enable us to have a more intentional relationship with technology and with each other. I am currently developing applications for capturing visual media that encourage people to take fewer shots and write notes about the captured content which can be used to organize and retrieve information without the need for computer vision or AI. My work with mobile AR collaboration is an example of creating technical "counterfoils" to headset-based communication, and there are many more contexts that could benefit from well-considered alternatives. There are many examples of community-led movements that encourage people to switch to more convivial infrastructures for communication, information-seeking, and record-keeping [25], and I will participate in and learn from these efforts to develop and demonstrate workflows that enable more people to take control of their interactions with technology using a combination of existing and newly-designed tool systems.

**2. Growing Counterfoil Communities:** There is a pressing need for a more critical practice in technology research, but there still is a divide between the "craft work of design" and the "reflexive work of critique" required for it to become a "praxis of daily work" [26]. As someone trained in both engineering and design, and who is working to develop a more critical understanding and the vocabulary that comes with it, I am very motivated to help people from both sides of the design-critique divide bridge this gap in their own practice [27]. I believe the best way to counter the top-down imposition of technology-driven control is a bottom-up approach of building a community of critical designers who can *think* of alternate visions of a convivial world, *turn* to work with communities they care about, and *try* to bring these visions into reality.

The world will continue to change, and the chance to shape that change for good as part of the community at the Keller Center—one that already embodies the Counterfoil spirit—would be, for me, the perfect next step.

## References


[1] Ivan Illich. *Tools for Conviviality*. Harper & Row, 1973.

[2] Jason Hickel. *Less is More*. Penguin Random House, 2020.

[3] Ivan Illich. *Toward a History of Needs*. Pantheon Books, 1978.

[4] Isabelle Stengers. “The Insistence of Possibles.” In: *PARSE Journal* (2017). Ed. by Anders Hultqvist, Dave Beech, and Valérie Pihet. DOI: `10.70733/w8eviuefu210`.

[5] Terry Winograd and Fernando Flores. *Understanding Computers and Cognition: A New Foundation for Design*. Ablex Publishing Corporation, 1986.

[6] *The Facebook Company Is Now Meta | Meta*. July 2026. URL: `https://web.archive.org/web/20260724051039/https://about.fb.com/news/2021/10/facebook-company-is-now-meta/`.

[7] Rishi Vanukuru. “Designing the Everyday Spatial Computer: Supporting Meaningful Collaboration using Mobile Devices.” PhD thesis. University of Colorado Boulder, 2026. URL: `https://rishivanukuru.com/dissertation2026`.

[8] Rishi Vanukuru, Suibi Che-Chuan Weng, Krithik Ranjan, Torin Hopkins, Amy Banic, Mark D. Gross, and Ellen Yi-Luen Do. “DualStream: Spatially Sharing Selves and Surroundings using Mobile Devices and Augmented Reality.” In: *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. Oct. 2023, pp. 138–147. DOI: `10.1109/ISMAR59233.2023.00028`.

[9] Rishi Vanukuru, Jens Emil Grønbæk, and Ellen Yi-Luen Do. *Designing Interfaces that Support Hybrid Work Across Seams with Mobile Spatial Computing*. In Preparation.

[10] Steve Benford, Gabriella Giannachi, Boriana Koleva, and Tom Rodden. “From interaction to trajectories: designing coherent journeys through user experiences.” In: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. CHI ’09. New York, NY, USA: Association for Computing Machinery, Apr. 2009, pp. 709–718. DOI: `10.1145/1518701.1518812`.

[11] Matthew Chalmers and Areti Galani. “Seamful interweaving: heterogeneity in the theory and design of interactive systems.” In: *Proceedings of the 5th conference on Designing interactive systems: processes, practices, methods, and techniques*. DIS ’04. New York, NY, USA: Association for Computing Machinery, Aug. 2004, pp. 243–252. DOI: `10.1145/1013115.1013149`.

[12] Janet Vertesi. “Seamful Spaces: Heterogeneous Infrastructures in Interaction.” In: *Science, Technology, & Human Values* 39.2 (Mar. 2014), pp. 264–284. DOI: `10.1177/0162243913516012`.

[13] Rishi Vanukuru, Krithik Ranjan, Ada Yi Zhao, David Lindero, Gunilla H. Berndtsson, Gregoire Phillips, Amy Banić, Mark D. Gross, and Ellen Yi-Luen Do. “Studying Mobile Spatial Collaboration across Video Calls and Augmented Reality.” In: *Proc. ACM Hum.-Comput. Interact.* 10.2 (May 2026). DOI: `10.1145/3788073`.

[14] Tyler Blackman and Daniel Harley. “Interpreting Apple’s visions: Examining the spatiality of the Apple Vision Pro.” In: *Platforms & Society* 1 (Nov. 2024). DOI: `10.1177/29768624241283913`.

[15] Chris Hesselbein. “The Neverending Story: Illusions of Limitlessness and Abundance in the Metaverse.” In: *Funes. Journal of narratives and social sciences* 8 (Dec. 2025), pp. 4–16. DOI: `10.6093/2532-6732/12786`.

[16] Sarah Kaplan and Wanda J. Orlikowski. “Temporal Work in Strategy Making.” In: *Organization Science* 24.4 (2013), pp. 965–995. URL: https://www.jstor.org/stable/42002889.

[17] Rishi Vanukuru, Payod Panda, Xinyue Chen, Ava Elizabeth Scott, Lev Tankelevitch, and Sean Rintel. “Designing Interfaces that Support Temporal Work Across Meetings with Generative AI.” In: *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. DIS ’25. New York, NY, USA: Association for Computing Machinery, July 2025, pp. 3600–3620. DOI: 10.1145/3715336.3735833.

[18] Janet Vertesi, danah boyd, Alex S Taylor, and Benjamin Shestakofsky. “Reckoning with the Political Economy of AI: Avoiding Decoys in Pursuit of Accountability.” In: *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency*. FAccT ’26. New York, NY, USA: Association for Computing Machinery, June 2026, pp. 2186–2205. DOI: 10.1145/3805689.3806739.

[19] Dan McQuillan. *Resisting AI: An Anti-fascist Approach to Artificial Intelligence*. Bristol University Press, 2022.

[20] Hagen Blix and Ingeborg Glimmer. *Why we fear AI: on the interpretation of nightmares*. Brooklyn, NY Philadelphia, PA: Common Notions, 2025.

[21] Emily M. Bender and Alex Hanna. *The AI Con: How To Fight Big Tech’s Hype and Create the Future We Want*. Harper Collins, 2025.

[22] Langdon Winner. “Do artifacts have politics?” In: *Deadalus* 109 (1) (1980), pp. 121–136.

[23] Sasha Costanza-Chock. *Design Justice: Community-Led Practices to Build the Worlds We Need*. The MIT Press, 2020.

[24] Christoph Becker. *Insolvent: How to Reorient Computing for Just Sustainability*. The MIT Press, 2023.

[25] *Reclaim Control | over your technology, your devices, your data, your digital life.* URL: www.techreclaimers.club.

[26] Phil Agre. “Toward a Critical Technical Practice: Lessons Learned in Trying to Reform AI.” In: *Bridging the Great Divide: Social Science, Technical Systems, and Cooperative Work*. Erlbaum, 1997.

[27] Maya Malik and Momin M. Malik. “Critical Technical Awakenings.” In: *Journal of Social Computing* 2.4 (Dec. 2021), pp. 365–384. DOI: 10.23919/JSC.2021.0035.